KUKU TV | AI GENERALIST ASSIGNMENT SUBMISSION

PROBLEM 1: AUTOMATED PROMO MINING & RANKIN

End-to-End Vertical Microdrama Ad Extraction across 50 Episodes

Applicant: Candidate (AI Generalist)
Roll No: ES23BTECH11021
Branch: AI & Data Engineering
Date: November 2026

* 1. PROBLEM CHOICE, FRAMING & CORE ASSUMPTIONS

Kuku's microdrama growth relies on high-velocity user acquisition (UA) on TikTok, Instagram Reels, and YouTube Shorts. Today, manual promo curation by human video editors creates a severe bottleneck as new shows release weekly. We built an automated, production-ready Promo Mining Engine that digests screenplay scripts, scores dramatic beats, and extracts high-converting 20-60s ad clips optimized for cold-traffic acquisition.

- **Ruthless 3-Second Rule:** Mobile viewers decide to swipe within 2-3s. A promo MUST start on an immediate shock (slap, insult, betrayal, reveal).
- **Zero-Context Intelligibility:** A cold viewer knows nothing about the lore. Promos must exploit universal, archetypal conflicts (cruel sister, disowned bride).
- **Unresolved Cliffhanger:** Promos must cut at peak psychological tension (impending blow, unanswered question) before any resolution occurs.
- **Duration Sweet Spot:** Strictly 20s-60s duration window, with maximal algorithmic retention score calibrated between 32s and 48s.

* 2. SYSTEM ARCHITECTURE: HYBRID HEURISTIC + NLP ENGINE

Rather than passing thousands of lines of raw text to expensive LLMs with large context windows, we engineered a deterministic, two-stage extraction pipeline that guarantees millisecond execution, zero hallucination, and full explainability:

Stage 1: Screenplay Parser

- Regex tokenization into discrete beats
- Extracts Start/End timestamps (MM:SS.SS)
- Segregates Camera, Action, Dialogue, Tone

Stage 2: Beat Scorer & Windowing

- Hook Scorer (0-3.5s): shock, physical, reveal
- Cliffhanger Scorer (T-3.5s to T): open loops
- Sliding window: 20s <= duration <= 60s

Stage 3: NMS & Global Ranker

- Non-Maximal Suppression (NMS, IoU > 12s)
- Composite Score: 0.35H + 0.35C + 0.2Cl + 0.1D
- Cross-series ranking & deduplication

* 3. RUN RESULTS & TOP-RANKED PROMOS ACROSS ALL 50 EPISODES

The engine parsed all 50 episodes of 'I Married a Broken Soldier. He was a Duke' (3,727 beats) and extracted 73 high-converting promo candidates. Below are the top ranked candidates ready for ad deployment:

Rank	Ep	Timestamps	Dur	Score	Hook Category	Cliffhanger Type
#1	Ep 23	01:19.00 - 02:11.50	52.5s	0.911	Physical / Poison Reveal	Ominous Cut to Black
#2	Ep 38	00:45.50 - 01:27.91	42.4s	0.864	Physical Peril / Cry	Ominous Cut to Black
#3	Ep 26	01:02.60 - 02:00.88	58.3s	0.834	Physical Confrontation	Ominous Cut to Black
#4	Ep 44	00:08.60 - 00:34.00	25.4s	0.817	Verbal Humiliation	Sudden Climax Action Cut
#5	Ep 12	00:43.70 - 01:27.21	43.5s	0.812	Physical / Blackthorn Threat	Impending Peril Cut
#8	Ep 3	01:31.70 - 02:25.75	54.1s	0.787	Swords Drawn at Altar	Mysterious Warrior Save
#14	Ep 1	01:47.50 - 02:17.50	30.0s	0.741	Altar Betrayal / 'Low Value'	Ominous Defiance / Black

PROBLEM 1: CAMPAIGN STRATEGY, SCALE & EVALU

Deployment Strategy, Scaling Economics, Failure Modes & Quality Checks

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* 4. CAMPAIGN STRATEGY: WHICH PROMOS TO POST FIRST AND WHY?

A naive algorithm posts the highest math score first. However, a growth marketer optimizes for the User Acquisition (UA) funnel. We recommend a 2-stage staged ad rollout strategy:

- **Campaign Phase 1 (Cold Traffic - Post First!):** Deploy Ep 1 (01:47.50-02:17.50) and Ep 12 (00:43.70-01:27.21). Why? The 'Public Betrayal at Altar' and 'Sister forcing bride to crawl through thorns' are universal human grievances. Zero backstory is needed; scroll-stopping emotional indignation drives the lowest Cost Per Install (CPI).
- **Campaign Phase 2 (Retargeting & Coin Unlock):** Deploy Ep 23 (01:19.00-02:11.50) and Ep 3 (01:31.70-02:25.75). Why? Once a viewer is hooked on the basic premise, the 'Someone is poisoning her' mystery and the 'Hidden Duke warrior reveals combat prowess' drive high-intent app opens and coin purchases.

* 5. HONEST QUALITY CHECK: WHERE DOES THE TOOL PICK BADLY?

Being transparent about boundary conditions is critical for real-world reliability. We identified three specific failure modes:

- **1. Visual-Only Shock Without Dialogue:** If an actor delivers a silent slap or dramatic eye expression documented solely in CAMERA notes, the text-based parser assigns lower emotional weight than a dialogue-heavy argument, under-scoring visceral visual moments.
- **2. Context-Dependent Pronouns:** In mid-series episodes (e.g. Ep 38, 'What if I die here?'), characters refer to past conspiracies using 'she' or 'the plan'. While tension is high, a cold TikTok user lacks context on who 'she' is, reducing conversion.
- **3. False Cliffhangers on Pacing Lulls:** Scripts often insert 'CAMERA: Cut to black' after quiet transitional scenes. Without cross-checking conflict density, a naive rule can misclassify a mundane exit as an urgent cliffhanger. (Mitigated by our conflict-density threshold).

* 6. REAL-WORLD SCALING ECONOMICS & HUMAN-IN-THE-LOOP

What We Gave to AI vs. Code

- Code: Regex tokenization, sliding time windows, non-maximal suppression
- AI/NLP: Emotional tone categorization, semantic conflict classification, pun
- Rationale: Keeps execution deterministic, 100% reproducible, and zero hal

Human-in-the-Loop & Production Scale

- Human Role: UA editor reviews top-5 candidates in the web UI and approves A/B tes
- Processing Speed: 50 episodes in 1.2s on a single CPU core; 100 full shows in <3 m
- Production Cost: $0.00 API cost for rule engine, <$0.02/show if enriched with LLM re

* 7. AI TOOLS USED & ONE PLACE THEY GOT IT WRONG

Tools Used: Gemini 1.5 Pro and Claude 3.5 Sonnet were used during early prototyping to explore prompt-based candidate extraction and screenplay structure analysis.

Where AI Got It Wrong: When prompted with raw screenplays to 'find the best promos', the LLMs repeatedly selected complete, resolved 90-120 second narrative arcs (e.g. including the setup, argument, AND the hero solving the issue). They instinctively tried to tell a satisfying, complete story rather than cutting abruptly on a high-stakes, unresolved cliffhanger. We solved this by enforcing hard programmatic sliding-window constraints (20-60s) and explicitly penalizing conflict resolution keywords.